

Risk sharing rules in modern heterogeneous tontines

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Outline

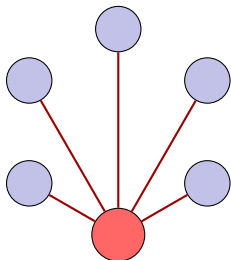
- 1 Introduction to Risk sharing
- 2 Setting and notation
- 3 The sharing mechanism

A real example : the Dutch *broodfonds* (bread fund)

- 73% of Dutch self-employed have *no* sickness cover
- Private insurance : too expensive

Their solution :

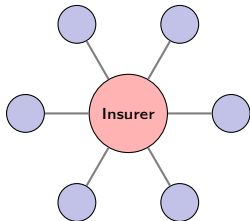
- ≈ 50 people form a fund
- Each keeps a personal account
- Contribution : €33–112 / month
- If you fall sick $\rightarrow$ the others pay you *from their own accounts*



sick member receives from all

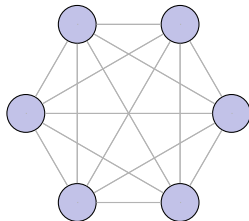
Centralized vs. decentralized

Centralized (insurance)



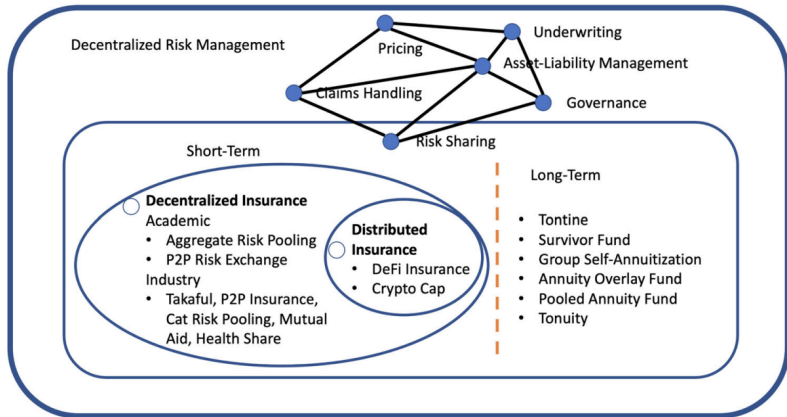
Risk *transferred* to a central bearer.
Capital, solvency, regulation, margins.

Decentralized (sharing)



Risk *shared* among peers.
No capital. No guarantee.

Figure – An overview of decentralized insurance and annuities.



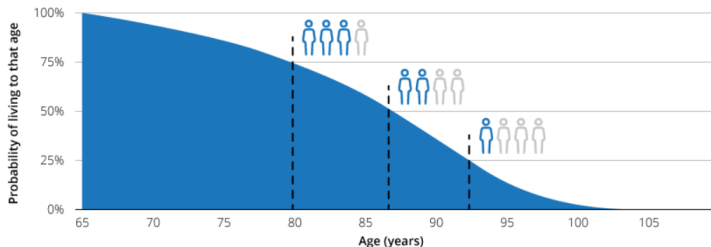
Scheme	Where
Online mutual aid	China, East Asia
Takaful	Middle East
Peer-to-peer insurance	Lemonade, Friendsurance
Catastrophe risk pools	African / Caribbean states
Retirement	Le Conservateur (FR), CREF (US), VPLA (Canada), PEPP (EU)

Figure – Longevity Risk (Retraite Québec)

Today, a man age 65 has:

- 3 out of 4 chances of reaching age 80
- 1 out of 2 chance of reaching age 86 (life expectancy)
- 1 out of 4 chance of reaching age 92.

Survival probability of a man age 65 until the age shown



Proportions de différents groupes d'âge, Québec, 1971 à 2071

	Total	0-19 ans	20-64 ans	65 ans et plus	55-64 ans	65-74 ans	75-84 ans	85 ans et plus	
	%								
1971	100,0	39,7	53,5	6,8	7,6	4,5	1,9	0,4	▲
1976	100,0	35,3	57,1	7,6	8,0	5,1	2,1	0,5	
1981	100,0	31,1	60,1	8,8	8,6	5,7	2,5	0,6	
1986	100,0	27,5	62,7	9,8	9,3	6,1	3,0	0,7	
1991	100,0	26,4	62,6	11,1	9,1	6,8	3,4	0,9	
1996	100,0	25,9	62,1	12,0	9,1	7,2	3,7	1,1	
2001	100,0	24,0	63,0	13,0	10,4	7,4	4,3	1,3	
2006	100,0	22,6	63,4	13,9	12,3	7,5	4,9	1,5	
2011	100,0	21,6	62,7	15,7	13,5	8,7	5,1	1,9	
2016	100,0	20,9	61,1	18,0	14,7	10,4	5,3	2,3	
2021	100,0	20,9	58,9	20,2	14,5	11,5	6,2	2,4	
2026	100,0	20,6	57,0	22,4	12,6	12,3	7,4	2,6	
2031	100,0	19,8	55,5	24,7	11,4	12,7	8,7	3,4	
2036	100,0	18,9	55,8	25,3	12,0	11,4	9,7	4,2	
2041	100,0	18,4	56,1	25,5	12,4	10,5	10,1	5,0	
2046	100,0	18,3	55,7	26,0	12,4	11,1	9,1	5,8	
2051	100,0	18,5	55,3	26,2	12,6	11,6	8,5	6,1	
2056	100,0	18,6	54,9	26,6	12,1	11,6	9,2	5,8	
2061	100,0	18,5	54,4	27,1	11,4	11,9	9,6	5,6	
2066	100,0	18,2	54,7	27,1	12,1	11,4	9,7	6,0	
2071	100,0	17,9	55,1	27,0	12,9	10,8	10,0	6,3	▼

Sources

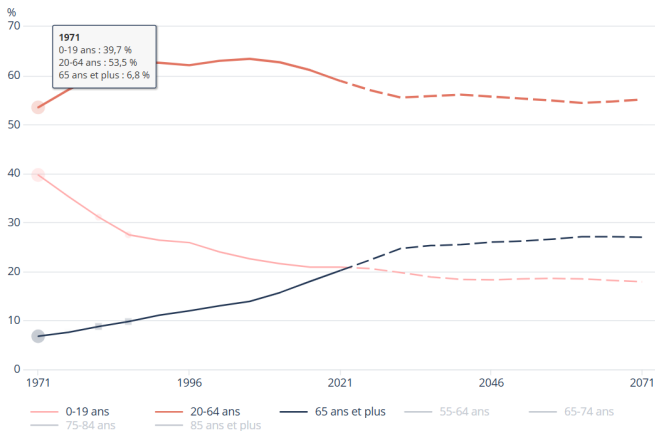
1971 à 2021 (données observées) : Statistique Canada, Estimations de la population (septembre 2024). Adaptation par l'Institut de la statistique du Québec.

2026 à 2071 (données projetées) : Institut de la statistique du Québec, Perspectives démographiques du Québec et de ses régions, mise à jour 2026, scénario de référence.

Mise à jour : 9 juillet 2026

Proportions de différents groupes d'âge, Québec, 1971 à 2071

Données observées : 1971-2021; données projetées : 2026-2071.



Sources

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- M homogeneous groups
- Group $m \in \{1, \dots, M\}$ has N_m members¹
- $N = \sum_{m=1}^M N_m$ is the total number of members in the longevity pooling plan
- each member k of group m contributes an amount c_m at time $t = 0$ which accumulates to $a_m = c_m(1 + r_m)$.²

1. copies of each other

2. For the rest of this presentation, we will assume $r_m = 0 \forall m$. the methodology stays the same for fixed or stochastic r_m as long as the investment risk and the mortality risk are independent.

Let $I_{m,k}$ denote the survival indicator of member k in group m ,

$$I_{m,k} = \begin{cases} 1 & \text{if the member survives the period,} \\ 0 & \text{otherwise} \end{cases}$$

with $\mathbb{P}(I_{m,k} = 1) = p_m = 1 - q_m$. We assume that $I_{m,k}$ are independent $\forall m \in \{1, \dots, M\}$, $k \in \{1, \dots, N_m\}$

- The number of survivors and deaths over the period in group

m are given by $\Lambda_m = \sum_{k=1}^{N_m} I_{m,k} \sim \text{Binomial}(N_m, 1 - q_m)$ and $D_m = N_m - \Lambda_m \sim \text{Binomial}(N_m, q_m)$ respectively

→ Longevity Credits :

- Let $X_{m,k} = a_m(1 - I_{m,k})$ denote the exposure (account value lost on death) of the member k in group m .
- the longevity credits available at the end of the period for the group m are $X_m = \sum_{k=1}^{N_m} X_{m,k} = D_m a_m$.
- the total longevity credits available in the scheme at the end of the period are given by $S = \sum_{m=1}^M X_m = \sum_{m=1}^M \sum_{k=1}^{N_m} X_{m,k}$.
- Note that S follows a Poisson binomial distribution.
- $\mathbb{E}[S] = \sum_{m=1}^M N_m q_m a_m$.
- Let $Z = S - \mathbb{E}[S]$ denote the aggregate deviation (realized mortality against expected mortality)³

3. Note that we do not consider the investment risk for now

In order to share the available longevity credits S between different members in the M groups, we use a two step approach :

- Step 1 : The Sharing rule acts between groups, allocating S into group credits $G_1, \dots, G_M$.
- Step 2 : Within each group $m \in \{1, \dots, M\}$, the group credit G_m is allocated uniformly and each member k receives the longevity credit $H_{m,k} = \frac{G_m}{N_m}$.
- The wealth of the member k in group m at the end of the period is $W_{m,k} = a_m I_{m,k} + H_{m,k}$
- Let $H_M = \left\{ (Y_1, \dots, Y_M) \in \mathbb{Y} : \sum_{m=1}^M Y_m = S \right\}$ and $H_M^{AF} = \left\{ (Y_1, \dots, Y_M) \in H_M : \mathbb{E}[Y_m] = \mathbb{E}[X_m] \forall m \right\}$

- Let $(Y_1^{(1)}, \dots, Y_M^{(1)}), \dots, (Y_1^{(K)}, \dots, Y_M^{(K)})$ denote K sharing-rules from H_M^{AF} and $Y_m^\delta = \sum_{k=1}^K \delta_k Y_m^{(k)}$ with $\sum_{k=1}^K \delta_k = 1$, then $(Y_1^\delta, \dots, Y_M^\delta) \in H_M^{AF}$
- The wealth of member k of group m is given by

$$W_{m,k} = a_m l_{m,k} + \frac{Y_m^\delta}{N_m}$$
- Note that

$$\begin{aligned} \sum_{m=1}^M \sum_{k=1}^{N_m} W_{m,k} &= \sum_{m=1}^M \sum_{k=1}^{N_m} (a_m - X_{m,k} + \frac{Y_m^\delta}{N_m}) \\ &= \sum_{m=1}^M \sum_{k=1}^{N_m} a_m \quad (\text{Full Allocation}) \end{aligned}$$
- $\mathbb{E}[W_{m,k}] = \mathbb{E}[a_m l_{m,k}] + \frac{1}{N_m} \mathbb{E}[X_m] = \mathbb{E}[a_m l_{m,k}] + \mathbb{E}[X_{m,k}] = a_m$

4

4. When we consider stochastic investment then the expected wealth at the end of the period is the expected financial gain

Examples of some simple sharing rules :

- Conditional mean risk sharing rule (CMRS) $Y_m = \mathbb{E}[X_m | S]$
- Proportional rule $Y_m = \frac{\mathbb{E}[X_m]}{\mathbb{E}[S]} S$
- regression rule $Y_m = \mathbb{E}[X_m] + \frac{\text{Cov}(X_m, S)}{\text{Var}(S)} S$
- Full retention within the group $Y_m = X_m$

The objective is a state-contingent claim.

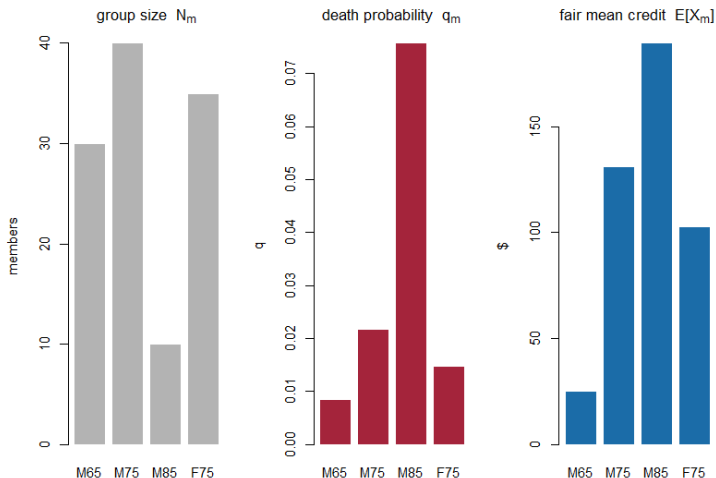
- The individual problem is

$$V_{m,k}(\delta) = \arg \max_{\delta \in \text{Simplex}} \mathbb{E}[U_{m,k}(\delta)] = \\ \mathbb{E}[U_m(a_m + \frac{Y_m^\delta}{N_m})I_{m,k} + \beta_m U_m(\frac{Y_m^\delta}{N_m})(1 - I_{m,k})].$$

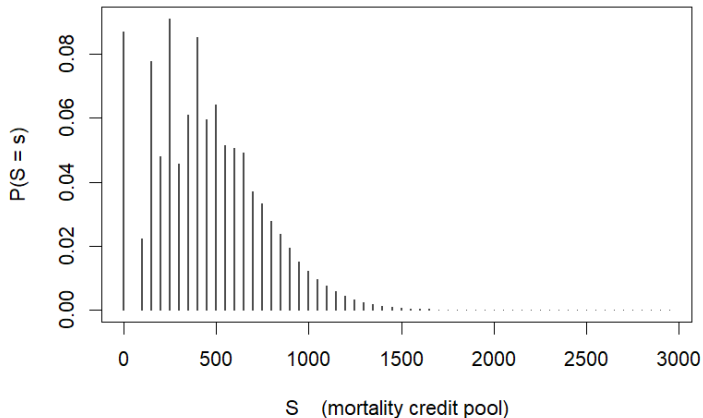
- The collective problem is

$$V(\delta) = \arg \max_{\delta \in \text{Simplex}} \sum_{m,k} V_{m,k}(\delta)$$

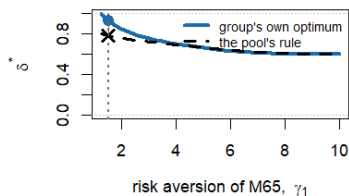
Figure – Using Canadian Pensioners' Mortality 2014 – Tables 1 CPM 2014 Rates.



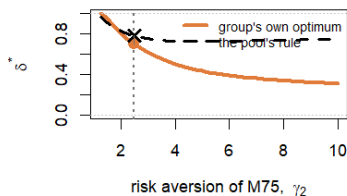
Exact distribution of the credit pool S



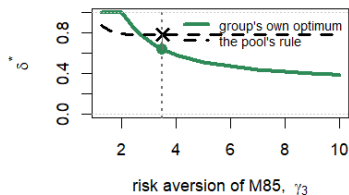
M65 ($q = 0.0084$, $s = 100$, $N = 30$)



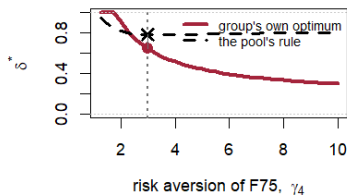
M75 ($q = 0.0218$, $s = 150$, $N = 40$)



M85 ($q = 0.0757$, $s = 250$, $N = 10$)

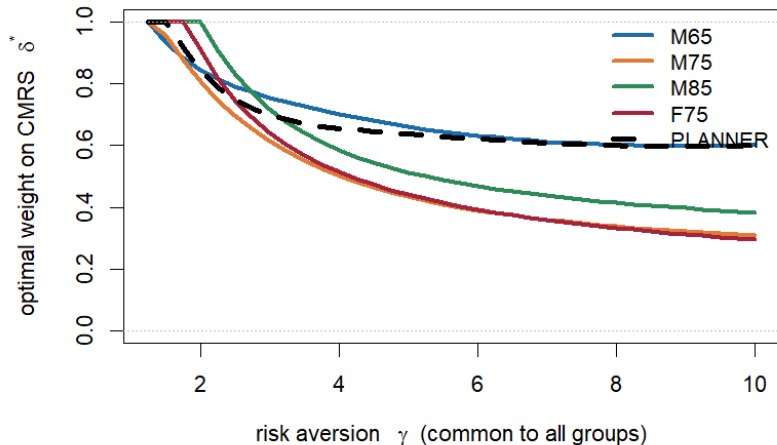


F75 ($q = 0.0147$, $s = 200$, $N = 35$)

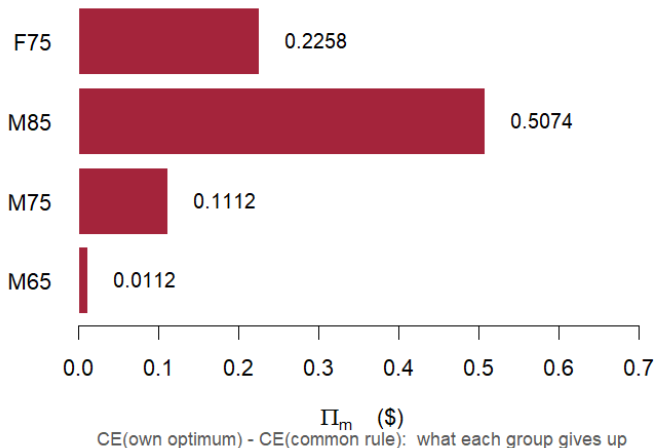


Optimal blend: each group vs the planner (beta = 0.50)

$\delta = 1$: pure CMRS $\delta = 0$: pure proportional



Cost of consensus (the pool posts $\Delta = 0.783$)



- Different non-connected employers in VPLA and CDC schemes.
- Smoothing guarantees chosen by each group m .
- Optimal stochastic investment portfolio of each group m in a Black-Scholes market.
- Secondary Market of Longevity credit exchange inside the scheme to offset the differences in the global chosen rule